

S3 ICT

Comprehensive Study Notes

St. Margaret's Co-educational School · Secondary 3

■ Python Programming

■ MIT App Inventor

■ Artificial Intelligence

■ Health & Ethics of ICT

■ Exam Revision Tips

Covers all exam topics: Python coding tasks, App Inventor classifier project, AI reasoning & machine learning, and ICT health & ethics.

Section 1: Python Programming

Data types · Operators · Conditions · Loops · Lists · Functions · Errors

1.1 What is Python?

Python is a popular, beginner-friendly programming language used in data science, web development, and Artificial Intelligence. In S3 ICT, we use Python to learn the fundamental concepts of programming.

Key facts: Python uses **indentation** (spaces) instead of brackets to organise code. Every block of code inside an *if*, *for*, *while*, or *def* must be indented by **4 spaces**. Missing or wrong indentation causes an **IndentationError**.

1.2 Data Types

Every piece of data in Python has a **type**. The type controls what you can do with it — for example, you can do maths with numbers but not with text.

Type	What it is	Examples
int	Whole numbers — no decimal point	5, -3, 0, 100
float	Numbers with a decimal point	3.14, -0.5, 2.0
str	Text — always in quotes	"hello", 'abc', "123"
bool	True or False — only two values	True, False

■ Useful built-in functions:

Function	What it does	Example
type(x)	Tells you the data type of x	type(5) → int (the class name)
int(x)	Converts x to an integer	int("42") → 42
float(x)	Converts x to a float	float("3.14") → 3.14
str(x)	Converts x to a string	str(99) → "99"
input(msg)	Asks the user to type something	name = input("Your name: ")
print(x)	Displays x on screen	print("Hello!")

■ **Type Casting Warning:** input() always returns a string. If you need a number, you must convert it: age = int(input("Age: "))
Forgetting this is a very common mistake!

1.3 Operators

Operators are symbols that perform operations on values. There are four main groups.

Arithmetic Operators (maths):

Symbol	Meaning	Example
+	Addition	5 + 3 = 8
-	Subtraction	10 - 4 = 6
*	Multiplication	3 * 4 = 12

/	Division	10 / 4 = 2.5 (always float)
**	Exponent (power)	2 ** 3 = 8
%	Modulo (remainder)	10 % 3 = 1
//	Floor division	10 // 3 = 3 (rounds down, gives int)

Comparison Operators (give True or False):

Symbol	Meaning	Example
==	Equal to	5 == 5 → True
!=	Not equal to	5 != 3 → True
<	Less than	3 < 7 → True
>	Greater than	7 > 3 → True
<=	Less than or equal to	5 <= 5 → True
>=	Greater than or equal to	6 >= 7 → False

Logical Operators (combine conditions):

Operator	Meaning	Example
and	Both conditions must be True	age > 0 and age < 18 → True if 1-17
or	At least one condition is True	x == 0 or x == 1 → True if x is 0 or 1
not	Reverses the condition	not True → False

■ **Common mistake:** = is for *assigning* a value (x = 5). == is for *comparing* two values (x == 5). Never mix them up!

1.4 Strings — Concatenation and Immutability

A **string** is a sequence of characters in quotes. Strings are **immutable** — this means once a string is created, you cannot change individual characters directly.

```
# Correct: joining two strings
greeting = "Hello" + " " + "World"
print(greeting)  # Output: Hello World

# WRONG: joining string with integer causes TypeError!
age = 16
msg = "I am " + age  # TypeError!

# CORRECT FIX: convert int to string first
msg = "I am " + str(age)  # Works fine
print(msg)  # Output: I am 16
```

■ **Key rule:** When using + to join text, everything must be a string. Use **str()** to convert numbers. This prevents the dreaded **TypeError**.

1.5 Conditions — if / elif / else

Conditions let your program make decisions. The program checks a condition and runs different code depending on whether it is True or False.

```

score = 75
if score >= 80:
    print("Distinction")      # Runs if score is 80 or above
elif score >= 50:
    print("Pass")            # Runs if score is 50-79
else:
    print("Fail")            # Runs if score is below 50

```

■ Rules to remember:

- Always put a **colon :** after the condition (if score >= 80:)
- Code inside the block must be **indented 4 spaces**
- You can have **multiple elif** blocks — as many as you need
- The **else** block is optional — it catches everything that didn't match
- Conditions are checked **in order** — once one matches, the rest are skipped

1.6 Loops — for and while

Loops let you repeat a block of code multiple times without writing it out again and again. Python has two types of loops.

The for loop — repeat a set number of times:

```

# Print numbers 0 to 4
for i in range(5):
    print(i)
# Output: 0 1 2 3 4

# range(start, stop) — from start up to (but NOT including) stop
for i in range(1, 6):
    print(i)
# Output: 1 2 3 4 5

# range(start, stop, step) — count in steps
for i in range(0, 10, 2):
    print(i)
# Output: 0 2 4 6 8

```

range() form	What it produces	Example
range(n)	0, 1, 2, ... n-1	range(5) → 0,1,2,3,4
range(a, b)	a, a+1, ... b-1	range(2,6) → 2,3,4,5
range(a, b, s)	a, a+s, a+2s, ... < b	range(1,10,3) → 1,4,7

The while loop — repeat while a condition is True:

```

count = 0
while count < 3:
    print("Count is:", count)
    count = count + 1    # IMPORTANT: update the variable!
# Output: Count is: 0 / Count is: 1 / Count is: 2

```

■■ Infinite Loop Warning: If you forget to update the variable inside a while loop, the condition will never become False, and the program will run forever. Always make sure something changes each time the loop runs!

1.7 Lists

A list stores **multiple items** in a single variable. Lists are **mutable** (you can change them). Each item has an **index** starting from 0.

```
fruits = ["apple", "banana", "cherry"]
print(fruits[0]) # Output: apple (index 0 = first item)
print(fruits[-1]) # Output: cherry (index -1 = last item)
print(len(fruits)) # Output: 3 (number of items)
fruits.append("mango") # Adds "mango" to the end
fruits.remove("banana") # Removes "banana"
fruits.insert(1, "kiwi") # Inserts "kiwi" at index 1
# Loop through a list
for fruit in fruits:
    print(fruit)
```

Operation	What it does
list[0]	Access first item (index 0)
list[-1]	Access last item
list.append(x)	Add x to the end of the list
list.remove(x)	Remove first occurrence of x
list.insert(i,x)	Insert x at position i
len(list)	Return number of items
list[i] = x	Change item at index i to x

1.8 Functions

A function is a named, reusable block of code. Instead of writing the same code again and again, you **define** it once and **call** it whenever you need it.

```
# Define a function
def greet(name):
    message = "Hello, " + name + "!"
    return message # Send the result back

# Call the function
result = greet("Kingsley")
print(result) # Output: Hello, Kingsley!

# Function with two parameters
def add(a, b):
    return a + b

total = add(3, 7)
print(total) # Output: 10
```

■ Variable naming rules:

- Must start with a **letter** (a-z, A-Z) or an **underscore** _
- Cannot start with a number (e.g. **1name** is invalid)
- Cannot contain spaces — use underscore: **my_name**
- Cannot be a Python keyword (e.g. **if**, **for**, **while**, **def**)

- Python is **case-sensitive**: **Name** and **name** are different variables

1.9 Common Python Errors

When something goes wrong, Python shows an **error message**. Learning to read error messages is an important skill — they tell you exactly what went wrong.

Error Type	Cause	Example
SyntaxError	Code is written incorrectly	Missing colon after if/for/while, mismatched brackets
IndentationError	Indentation is wrong or missing	Forgot to indent code inside an if block
TypeError	Wrong type used in an operation	Trying to do "hello" + 5 (str + int)
NameError	Variable used before it was defined	print(x) but x was never assigned a value
IndexError	Tried to access a list index out of range	list has 3 items, but you tried list[5]
ZeroDivisionError	Tried to divide by zero	10 / 0 causes this error
ValueError	Value is wrong type for an operation	int("hello") — cannot convert "hello" to int

Section 2: MIT App Inventor — Personal Image Classifier

Training a model · Components · Blocks logic · Improving accuracy

2.1 What is MIT App Inventor?

MIT App Inventor is a free, visual programming platform that lets you build real Android apps using **drag-and-drop blocks** — no text coding needed. In S3 ICT, we use it to build a **Personal Image Classifier** app that can recognise objects from the camera.

The platform has two views: the **Designer** (where you add UI components like buttons and labels) and the **Blocks editor** (where you write the logic using visual blocks).

2.2 The 6 Steps to Train Your Model

Before using the classifier in App Inventor, you must **train a model** using the Personal Image Classifier (PIC) website. This teaches the AI what each object looks like.

Step 1	Go to the PIC website	Visit the Personal Image Classifier tool (linked from the App Inventor website).
Step 2	Create a new project	Start a new project and give it a name related to your classification task (e.g. Dim Sum Classifier).
Step 3	Name your classes	Create a class for each category you want to recognise (e.g. 'Siu Mai', 'Har Gau', 'Char Siu Bao'). Each class is one type of object.
Step 4	Add training images	For each class, take or upload multiple photos. The more varied the photos (different angles, lighting, backgrounds), the better the model will perform. Aim for at least 20-30 images per class.
Step 5	Train the model	Click the 'Train' button. The AI analyses all your photos and learns the patterns that distinguish each class.
Step 6	Export and download	Test the model accuracy. If it performs well, export and download the model file (.mdl) to use inside App Inventor.

2.3 Required App Inventor Components

Components are the building blocks of your app. Some are **visible** (appear on screen) and others are **non-visible** (do work in the background, not seen by the user).

Visible Components (shown on screen):

Component	Purpose
WebView	Displays the live camera feed so the user can aim their phone at an object for classification. Without this, the classifier cannot see anything.

Button	The user taps this to trigger the classification. Usually labelled something like 'Classify' or 'What is this?'
Label	Displays the classification result as text on screen (e.g. 'This is Siu Mai — 92% confident').

Non-Visible Components (work in the background):

Component	Purpose
PersonallImageClassifier	The AI engine. It processes the camera image and returns the predicted class label and confidence score. You must upload your trained .mdl file here.
TextToSpeech	Reads the classification result out loud using the phone's speaker. This makes the app accessible to people who have difficulty reading.

2.4 Key Blocks Logic

In the Blocks editor, you connect visual blocks to create the logic of your app. Here is the typical flow for the classifier:

Block	What it does
When Button.Click	The event that fires when the user taps the Classify button
call PersonallImageClassifier.ClassifyVideoData	Sends the current camera frame to the AI model for analysis
When PersonallImageClassifier.GotClassification	The event that fires when the AI has finished analysing and has a result
set Label.Text to result	Displays the classification result (class name + confidence) in the Label
call TextToSpeech.Speak	Reads the result label out loud

2.5 Improving Model Accuracy

A model's **accuracy** is how often it correctly identifies an object. If your model works well in testing but fails in real life, the training data was not diverse enough.

- **More images per class:** Use at least 20-30 photos. More is better.
- **Vary the conditions:** Take photos from different angles, distances, and lighting conditions (bright, dim, natural, artificial).
- **Vary the background:** Don't always use the same tablecloth or surface. Real-world backgrounds are messy.
- **Balance your classes:** Have roughly the same number of photos for each class. If one class has 100 images and another has 5, the model will favour the bigger class.
- **Avoid overfitting:** If the model memorises the training photos instead of learning general patterns, it will fail on new photos. Diverse training data prevents this.
- **Test with new images:** After training, test with photos that were NOT used in training. This gives you a true measure of accuracy.

■ **Exam Tip:** If asked 'why does the model work in testing but fail in real use?' — the answer is that the training images were too similar / not diverse enough. This is called **overfitting**.

Section 3: Artificial Intelligence & Machine Learning

Types of reasoning · Machine learning · Future of work · Key vocabulary

3.1 What is Artificial Intelligence?

Artificial Intelligence (AI) is the ability of a computer or machine to perform tasks that would normally require human intelligence — such as recognising images, understanding speech, playing games, or making decisions.

AI is different from traditional programming. In normal programming, a human writes exact instructions for every situation. In AI, the machine **learns from data** and figures out the rules itself.

3.2 Three Types of AI Reasoning

AI robots can reason (think and make decisions) in three different ways, from the simplest to the most sophisticated:

1. Instinct-based reasoning

Fast, automatic reactions with NO conscious thought — like a reflex.

Example: A robot instantly moves backwards when it detects a wall. No thinking involved.

2. Experience-based reasoning

Actions based on PATTERNS learned from past experience or training data.

Example: A spam filter that has seen millions of emails learns to identify spam patterns.

3. Knowledge-based reasoning

Deep analysis using process knowledge, experience, and clear objectives. The most advanced type.

Example: A medical AI that analyses symptoms, uses medical knowledge, and recommends a diagnosis.

3.3 Three Types of Machine Learning

Machine Learning (ML) is the most common way to build AI today. Instead of programming rules by hand, we give the machine lots of data and let it learn patterns on its own. There are three main types:

Type	How it works	Real-world examples	Key feature
Supervised Learning (SL)	The AI learns from labelled data — pairs of inputs AND correct outputs provided by a human supervisor.	Personal Image Classifier (you label each class yourself). Spam filter trained on emails labelled 'spam' or 'not spam'.	<i>Most common type in practice</i>
Unsupervised Learning (UL)	The AI finds patterns in data WITHOUT any labels. It groups similar items together on its own.	Spotify grouping songs into playlists you didn't create. Customer segmentation in marketing.	<i>No labels needed — discovers hidden patterns</i>

Reinforcement Learning (RL)	The AI learns by trial and error — it receives rewards for good actions and penalties for bad ones.	AlphaGo learning to play chess. A robot learning to walk by trying different movements.	<i>Used for games, robotics, self-driving cars</i>
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3.4 AI and the Future of Work

AI is changing the job market. Some jobs are becoming **less in demand** because AI can do them faster and cheaper. Other jobs are becoming **more in demand** because humans are needed to build, manage, and supervise AI systems.

World Economic Forum (WEF) — 'Future of Jobs Report 2020' — Predictions by 2025:

Job	Demand	Reason
Data Entry Clerks	Decreasing	Routine data input is automated
Accounting/Bookkeeping Clerks	Decreasing	Software can handle routine accounts
Administrative Secretaries	Decreasing	Scheduling and admin tasks are automated
AI & Machine Learning Specialists	Increasing	Need people to build and train AI systems
Big Data Specialists	Increasing	Analysing large datasets requires expertise
Internet of Things (IoT) Specialists	Increasing	Connecting smart devices needs specialists
Digital Marketing Specialists	Increasing	Online business growth drives demand

OECD — 'Future of Education and Skills 2030' — Three skill classes students need:

Skill Class	What it includes
Cognitive & Meta-cognitive Skills	Critical thinking, creativity, problem-solving, self-regulation — knowing how to think and learn
Social & Emotional Skills	Empathy, teamwork, communication, leadership — working well with others
Practical & Physical Skills	Using tools and technology, physical coordination, hands-on tasks

■ **Exam Tip:** Make sure you can name all THREE skill classes from the OECD framework exactly. Also remember which WEF jobs are decreasing (data entry, accounting, admin) vs increasing (AI/ML, Big Data, IoT).

Section 4: Health & Ethics of ICT

Physical health · Mental health · Digital divide · Netiquette · Privacy & copyright

4.1 Physical Health Issues from ICT Use

Spending too long using computers, phones, and other ICT devices can cause real physical health problems. It is important to take regular breaks and maintain good posture.

Health Issue	Cause	Symptoms/Effects
Repetitive Strain Injury (RSI)	Damage to nerves, tendons, and muscles caused by making the same small movements over and over — e.g. typing, clicking a mouse.	Carpal Tunnel Syndrome (pain and numbness in the wrist and hand)
Text Neck	Neck and upper spine pain caused by tilting the head forward to look down at a phone or tablet for long periods.	Neck stiffness, shoulder pain, headaches
Eye Strain (Computer Vision Syndrome)	Tiredness, blurry vision, dry eyes, and headaches caused by staring at a bright screen for too long without breaks.	Blurred vision, sore eyes, headaches
Sedentary Lifestyle	Spending too many hours sitting still at a computer instead of moving. Long-term sitting increases risk of obesity and heart disease.	Weight gain, poor circulation, back pain

4.2 Mental Health Issues from ICT Use

ICT can also affect our mental and emotional wellbeing, especially when used excessively or irresponsibly.

- **Cyberbullying:** Harassment, threats, or humiliation carried out through digital devices and platforms. Can cause severe anxiety and depression.
- **Social Isolation:** Spending so much time online that face-to-face friendships deteriorate, leading to loneliness.
- **Internet/Screen Addiction:** Compulsive, uncontrollable use of the internet or social media, similar to other behavioural addictions.
- **Sleep Disruption:** The blue light emitted by screens suppresses melatonin production, making it harder to fall asleep.
- **FOMO (Fear of Missing Out):** Anxiety caused by seeing others' highlight reels on social media and feeling left out.
- **Anxiety and Low Self-Esteem:** Constant comparison with others on social media can damage self-image.

4.3 The Digital Divide & Equity of Access

Not everyone in the world has equal access to ICT. This creates serious inequalities in education, work, and social participation.

Global Digital Divide

The gap between people who have access to modern information and communication technology (computers, internet, smartphones) and those who do not. This divide exists between rich and poor countries, and also within countries between different socioeconomic groups.

Equity of Access

The principle that every person, regardless of their income, location, disability, age, or gender, should have an equal opportunity to use and benefit from ICT. It is not just about having a device, but also about having the skills, connectivity, and support to use it effectively.

Causes of the digital divide:

- **Cost:** Devices and internet access are expensive — many families cannot afford them
- **Infrastructure:** Rural or developing areas may lack reliable electricity or broadband
- **Digital literacy:** Older generations or those with less education may not know how to use technology
- **Language barriers:** Much online content is in English, disadvantaging non-English speakers
- **Disability:** Not all technology is accessible to people with visual, hearing, or motor impairments

4.4 Netiquette

Netiquette = **Network** + **etiquette**. These are the rules of polite, respectful behaviour when communicating online.

- **Do not type in ALL CAPS** — it is considered shouting and is seen as aggressive
- **Be respectful and kind** — treat others online as you would in person
- **Think before you post** — once something is online, it is very hard to take back
- **Protect others' privacy** — do not share personal photos or information about others without permission
- **Give credit** — cite sources and do not plagiarise others' work
- **Be careful with humour and sarcasm** — tone is difficult to convey in text and jokes can be misunderstood
- **Do not spam** — do not send large numbers of unsolicited messages

4.5 Privacy, Copyright & Security

Concept	Explanation
Copyright	A legal right that gives creators (authors, artists, programmers) exclusive control over how their work is used, copied, or shared.
Plagiarism	Using someone else's work (writing, code, images) as your own without giving credit. It is both unethical and often illegal.
Data Privacy	Personal information (name, address, health records) must only be collected and used with the person's consent.
Phishing	Fake emails or websites designed to trick users into revealing passwords, credit card numbers, or other sensitive data.
Strong Password	A password that is long (8+ characters), uses a mix of letters, numbers, and symbols, and is not reused across sites.
Two-Factor Auth (2FA)	A second verification step (e.g. a code sent to your phone) added on top of a password for extra security.

Section 5: Exam Revision & Quick Reference

Exam format · Must-memorise facts · Common mistakes · Quick reference tables

5.1 Exam Format Overview

Section	Type	Content
Quiz — Part A	Matching	Match Python code snippets to their descriptions (dropdown)
Quiz — Part B	Short Questions	App Inventor classifier steps, accuracy tips, TextToSpeech usage
Written — Part A	10 MCQs	Multiple choice across Python, AI, App Inventor, Health & Ethics
Written — Part B	Matching	Python code snippets matched to descriptions
Written — Part C	Fill in blank	Key terms and facts across all topics
Written — Part D	Structured Q's	App Inventor (6 marks), Python debug/write (8 marks), AI & Future of Work (6 marks)

5.2 Top 15 Facts to Memorise

Each card shows one key fact. The coloured strip on the left tells you which topic it belongs to. Read the bold heading first, then the detail below.

App Inventor 01	6 steps to train a model * PIC website * Create project * Name classes * Add training images * Click Train * Export / download .mdl
App Inventor 02	3 required components * WebViewer — shows the live camera feed * PersonalImageClassifier — the AI engine that analyses the image * TextToSpeech — reads the result aloud
Python 03	Strings are IMMUTABLE — Lists are MUTABLE * Strings: once created, individual characters CANNOT be changed directly * Lists: can be changed at any time (append, remove, insert, etc.)
Python 04	TypeError — joining string + integer * Cause: using + to join a str and an int without converting * Fix: use str() first e.g. "I am " + str(16)

Python 05	range() always starts from 0 * range(5) -> 0, 1, 2, 3, 4 (NOT 1 to 5!) * range(1, 6) -> 1, 2, 3, 4, 5 * range(0, 10, 2) -> 0, 2, 4, 6, 8
AI & ML 06	3 types of AI reasoning * Instinct-based: fast automatic reactions, no conscious thought * Experience-based: actions based on patterns learned from past data * Knowledge-based: deep analysis using knowledge + objectives (most advanced)
AI & ML 07	3 types of machine learning * Supervised: learns from labelled data (input + correct output provided) * Unsupervised: finds patterns in data with NO labels * Reinforcement: learns by trial & error using rewards and penalties
AI & ML 08	Personal Image Classifier = Supervised Learning * You provide the labels (class names) yourself when training * The model learns to match images to the labels you gave it
AI & ML 09	WEF 2020 — jobs with DECREASING demand by 2025 * Data Entry Clerks * Accounting / Bookkeeping Clerks * Administrative Secretaries
AI & ML 10	WEF 2020 — jobs with INCREASING demand by 2025 * AI & Machine Learning Specialists * Big Data Specialists * IoT (Internet of Things) Specialists
AI & ML 11	OECD 3 skill classes (Future of Education 2030) * Cognitive & Meta-cognitive — critical thinking, creativity, problem-solving * Social & Emotional — empathy, teamwork, communication, leadership * Practical & Physical — using tools, technology, hands-on tasks
Health & Ethics 12	Global Digital Divide * The gap between people who HAVE access to modern ICT * and people who DO NOT have access (internet, devices, digital skills)
Health & Ethics 13	Equity of Access * Everyone deserves equal opportunity to benefit from ICT * Regardless of income, location, age, disability, or gender

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- * Cause: repetitive small movements over a long period (e.g. typing, mouse clicks)
- * Damage: nerves and tendons in the hand/wrist
- * Example: Carpal Tunnel Syndrome

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- * Cause: tilting the head forward/downward to look at a phone or screen
- * Effect: pain and stiffness in the neck and upper spine

Mistake	Correct understanding	Example
Using = instead of ==	= assigns a value. == compares. Use == inside conditions.	if x = 5: is WRONG. if x == 5: is correct
Missing colon :	Every if/elif/else/for/while/def line MUST end with a colon.	if score >= 50 is WRONG. if score >= 50: is correct
Off-by-one with range()	range(5) gives 0,1,2,3,4 — NOT 1,2,3,4,5. Range stops BEFORE the end.	range(1,6) gives 1,2,3,4,5
Forgetting str() conversion	input() returns a string. If you need a number, use int() or float().	int(input("Enter age: "))
Supervised vs Unsupervised	Supervised = labelled data. Unsupervised = NO labels.	PIC uses supervised learning
RSI vs Text Neck	RSI = nerves/tendons in hands/wrists. Text Neck = neck/spine pain.	Different body parts, different causes
WebView vs WebViewer purpose	WebView shows the camera feed. PersonImageClassifier does the AI.	Don't confuse these two components

[illegible]

